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This Project Summary (PSD # 54561) is unofficial, English-machine-translated reference copy derived from public disclosure, published in the European Bank for Reconstruction and Development (EBRD) project repository.

Uzbekistan: Pilot Project production of "green" hydrogen

Location:

Uzbekistan

Project number:

54561

Industry:

energy

Sector:

private

Approval date:

March 6, 2024

State:

in development

Date of publication of the project summary in English:

January 29, 2024

Project Description

The project involves the allocation of a senior secured loan in the amount of up to \$58.2 million in project financing for the construction of a green hydrogen production complex with

using renewable resources, consisting of a 20 MW electrolyzer powered by a 58 MW wind farm with an annual production of 3,000 tons of "green" hydrogen. The electrolyzer will be located in the city of Chirchik in the Tashkent region, and the wind farm will be located in the Gijduvan district of the Bukhara region.

the installation will be located on the territory of the operating Bashskaya wind farm with a capacity of 500 MW, financed by the EBRD.

Project objectives

With the help of financing a green hydrogen production complex

The project aims to promote:

- a) the creation of the first production in Uzbekistan and Central Asia "green" hydrogen, which should become a clear example for development other similar projects;
- b) decarbonization of the ammonia production sector in Uzbekistan;
- c) expansion of electricity production from renewable sources.

Impact on the transient process

Expected Impact on Transition Score: 80

The project will contribute to the "green" transition process by establishing Uzbekistan's first green hydrogen production facility using electrolysis technology. The project aims to serve as a clear example for market development.

"green" hydrogen by attracting the private sector in Uzbekistan and the region as a whole.

The project is central to the Bank's overall financial management activities. dialogue on issues of public policy in the sphere of production low-carbon hydrogen and ammonia decarbonization industry in Uzbekistan.

Client Information

JV LLC "ACWA POWER UKS GREEN H2"

"SP LLC "ACWA Power UKS Green H2 LLC JV" (Borrower), newly created a special purpose vehicle registered in Uzbekistan for the development and implementation of project. The borrower is owned by ACWA Power Company and JSC UzKimyoSanoat (UKS).

ACWA Power is an international developer of power plant projects, desalination plants, and green hydrogen production.

UKS is a chemical manufacturer located in the Republic of Uzbekistan. One of UKS's divisions will act as the buyer.

"green" hydrogen produced within the framework of the project.

EBRD financing

58 million 200 thousand US dollars

Total cost of the project

95 million 400 thousand US dollars

Complementarity

Financing structure: EBRD complementarity is ensured by the allocation of long-term financing for this major decarbonization project on terms that are not available in the Uzbek market.

Changes in public policy, industry, institutional regulatory structures or mechanisms: the EBRD's participation in the project is considered to provide additionality as it directly contributes to the development of renewable energy (RE) and a green hydrogen certification mechanism. This mechanism not only allows for the creation of the currently absent market for VE certificates in the country, but also ensures the improvement of environmental practices and sustainability at the national level.

Environmental and social issues

The project is classified as category A (in accordance with the 2019 edition of the ESP)

on the basis that it involves the expansion of a wind farm (Bashskaya WPP) already financed by the Bank. The project also includes

construction on the territory of an existing ammonia plant a separate electrolysis plant that will produce "green" hydrogen using energy generated by additional facilities

wind power plants (after they are put into operation) on the basis of an agreement on compensation deliveries. The construction component of the project

electrolysis installations, classified as category B. In terms of capacity expansion An ESIA package for the Bashskaya Wind Power Plant 500 (52 MW) was prepared and released on December 21, 2023, for a 60-day public consultation period, after which the project will be submitted to the EBRD Board of Directors for review. An independent consultant and the Bank conducted an assessment. An ESIA was also conducted to obtain a construction permit. electrolyzer, where the Bank's environmental and social requirements were taken into account. This The ESIA was also reviewed by an independent consultant.

the design of a non-technical resume for the electrolyzer it will be published along with this project summary. For structuring

An ESAP is currently being prepared for the project in accordance with the EBRD's environmental and social requirements.

The Bashskaya wind power plant with an installed capacity of 500 MW (52772) is located in Bukhara region of Uzbekistan and is currently under construction. Due to its size and sensitive location

(biodiversity) and associated eco-social risks, it was classified as category A. The proposed project envisages increasing its capacity by 52 MW, which will require the installation of an additional 8 turbines of 6.5 MW each. The same applies to the additional capacity

Category A. ESIA package for additional capacities, prepared by independent consultants, includes the ESIA report, SEP, non-technical summary and environmental and social management system framework document management plan, including comprehensive biodiversity data and a cumulative impact

assessment. Other existing management documentation (including the Resettlement Action Plan) has also been updated.

Electricity from the additional capacity will be transmitted via an overhead power line running to the Bashskaya Wind Farm. The ESIA concluded

that additional wind power capacity will provide the same impact, as the current wind farm, and it can be regulated with the help of

already planned measures to minimize impacts.

A pilot electrolysis plant for hydrogen production will be located on the premises of the

UKS - MAXAM Chirchiq ammonia plant in Chirchik. Based on

Due to the size of the installation and the fact that its associated environmental and social risks and impacts are readily mitigated, it is classified as a Category

B project. The ESIA, as well as the E&S Management System Framework Document, the Non-Technical Summary, and the SEP were prepared by an independent consultant. The electrolyzer is to be designed using "best available techniques" (BAT) and have appropriate safety controls and their assessment. The ESIA states that the impacts associated with the electrolyzer are limited. The electrolyzer will consume small volumes of water from the ammonia plant's water supply system. This should not affect available water resources. Placement of the electrolyzer on the territory of an existing production facility, it could potentially increase industrial safety risks, which is exacerbated by the presence of housing

near the plant. The electrolyzer will produce a small amount

The amount of hydrogen (3,000 tons) will be fed directly into the ammonia plant system. No hydrogen storage is required. Safety measures for electrolyzer operation will need to be integrated into the plant's industrial safety assessment system and procedures, as well as emergency response measures. Under these conditions,

It is expected that risks to industrial safety will be sufficiently degrees are minimized.

Currently, the Bank, with the support of a consultant, is completing its a comprehensive survey and develops a project plan. Based on the results Following a comprehensive review, this project summary will be updated.

The ESIA package can be found at <https://www.ebrd.com/work-with-us/projects/esia/pilot-uzbek-green-hydrogen-project.html>.

Technical cooperation and grants financing

This project involves the implementation of technical tasks cooperation (TC) on the implementation of a methodology for certifying i) electricity generated at renewable energy facilities in the country as being obtained using renewable resources, and ii) the production of renewable fuels from non-biological raw materials (including “green” hydrogen and its derivatives) in

country. This TC assignment will be financed from the EBRD Shareholder Special Fund (SSF).

Contact person in the Company

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Information on project implementation

Date of last update of the project summary

January 29, 2024

What is a transient process?

Further information on the EBRD's approach to assessing the projected transition impact of projects is available [here](#).

Opportunities to participate in projects

For any questions regarding participation in the project or procurement, please contact the client company.

For non-procurement related cooperation with the EBRD, please contact: Tel.: +44 20 7338 7168

email email: projectenquiries@ebrd.com

Public sector projects: see [EBRD Procurement](#)

tel.: +44 20 7338 6794

email email: procurement@ebrd.com

General information

Specific information requests can be made using the [EBRD information request form](#).

Environmental and Social Policy (ESP)

[ESP](#) and its associated Performance Requirements (PRs) set out the process by which the EBRD implements its commitment to promoting "environmentally responsible and sustainable development." In the PRPs

The TRs and the PRs contain specific requirements for clients to comply with relevant national legislation on public information and consultation, as well as to establish a grievance mechanism to address and facilitate the resolution of stakeholder concerns and grievances, particularly regarding the client's and project's environmental and social performance. Depending on the nature and scale of a project's environmental and social risks and impacts, the EBRD additionally requires its clients, where appropriate, to disclose information about the risks and impacts of projects and to conduct meaningful stakeholder consultations to obtain and respond to their feedback.

Further information on the EBRD's practices in this area is contained in [the ESP](#).

Business integrity and compliance with established standards

The EBRD's Office of the Chief Compliance Officer (OCCO) promotes responsible governance and ensures that all aspects of the Bank's operations meet the highest standards of integrity and are consistent with international best practice. Integrity due diligence is conducted on all Bank clients to eliminate unacceptable risks associated with misconduct, as well as reputational risks to the EBRD. The Bank believes that identifying and addressing problematic issues at the project assessment and approval stages is the most effective means of ensuring the integrity of the Bank's transactions. The Office of the Chief Compliance Officer (OCCO) plays a key role in these preventative measures and also helps monitor risks to the Bank's business reputation at the post-investment stage.

The Compliance Control Department is also responsible for investigating allegations of fraud, corruption, and misconduct in EBRD-financed projects. Anyone within or outside the Bank who suspects fraud or corruption must notify the Head of Compliance in writing by emailing compliance@ebrd.com. All reported facts will be reviewed by the OCC for appropriate action. All reports, including anonymous ones, will be verified. These reports may be made in any language of the Bank or the Bank's country of operations. Such information must be provided in good faith.

Access to Information Policy (AIP)

The IDP, which came into force on 1 January 2020, sets out the EBRD's procedures for disclosure and consultation.

with stakeholders to promote greater knowledge and understanding of its strategies, policies, and operations. For available information, please visit the "[Access to Information Policy](#)" section. on the EBRD website.

Specific information requests can be made using the [EBRD information request form](#).

Independent Project Accountability Body (IPAB)

If attempts to address environmental, social or disclosure concerns about a client or the Bank (e.g. through a client-established project-level grievance mechanism or through direct engagement with Bank management) are unsuccessful, individuals and organisations may seek to address their concerns through the EBRD's Independent Project Accountability Mechanism (IPAM).

The NSPA independently assesses project issues that, according to complainants, have either already caused harm or are likely to cause harm in the future. The NSPA's responsibilities include: maintaining dialogue between project stakeholders to resolve environmental, social, and disclosure issues; assessing the Bank's compliance with its [Environmental and Social Policy](#), as well as the provisions of [the Access to Information Policy](#) related to the draft ; and, if necessary, eliminating existing cases of non-compliance with these regulatory documents, taking measures to prevent the Bank from non-compliance with them in the future.

Visit [the Independent Project Accountability Authority website](#), To learn more about the NOPP and its mandate, and to find out how [to apply](#) to conduct an inspection, or [contact](#) the NPPP by email [at ipam@ebrd.com](mailto:ipam@ebrd.com) with questions, for additional information or [application instructions](#).